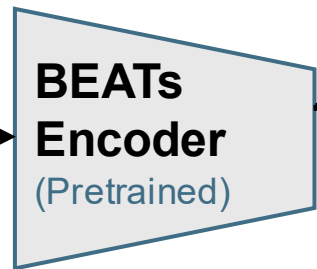


Respiratory Audio



*: Training + Inference
#: Inference Only



H_i

$T_i \times D$

Mean Pooling
(Linear 768→256)

Attention $\mathcal{G}$
(PAFA projection)

1×768 Z_i

Linear Heads

A: Abnormal

C: Crackle

W: Wheeze

p_i
(Shared Attributes)

HF C/W supervision

Task-Specific Outputs

ICBHI*:
Normal · Crackle · Wheeze · Both

SPRSound*:
Normal · Adventitious(Abnormal)

HF source-test#:
CAS

KAUH#:
Normal · Abnormal

$$\mathcal{L} = \underbrace{\mathcal{L}_{\text{cls}} + \lambda_p \mathcal{L}_{\text{PCSL}} + \lambda_g \mathcal{L}_{\text{GPAL}}}_{\mathcal{L}_{\text{core}}} + \lambda_{\text{HF}} \mathcal{L}_{\text{HF}}$$

Classification

Patient Cohesion–Separation

Global Patient Alignment

HF Lung Loss (Optional C/W)